

CredResolve Collection Intelligence

Data Analyst Assignment

Collections Analytics & Prioritization

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Tools: Python | SQL | Statistics | Data Visualization

Executive Summary

CredResolve operates in technology-driven recovery management, where effective collection depends on identifying the right accounts, selecting appropriate collection channels, and converting collection efforts into successful payment outcomes.

This project analyzes the provided multi-table collections dataset to understand portfolio exposure, collection activity, promises-to-pay, payment behaviour, delinquency, and data-quality issues.

The analysis follows the complete collection journey:

Account → Targeting → Collection Activity → Engagement → Promise to Pay → Payment

The portfolio contains approximately **₹10.49 billion in outstanding exposure** and **₹1.88 billion in recorded payments**, resulting in an overall collection rate of **17.92%**. There are **13,532 accounts with recorded promises-to-pay** and **16,934 accounts with recorded payments**.

Before deriving these metrics, the underlying datasets were assessed for missing values, duplicates, identifier consistency, and relationships between tables. The analysis then used Python, SQL, statistical measures, and visualizations to identify collection patterns.

Finally, an explainable collection-prioritization framework was developed using financial exposure, delinquency, payment activity, and unresolved promises-to-pay. The objective is to help collection teams focus their efforts on accounts where recovery attention may have the greatest business impact.

Then make a small KPI box:

KPI	Value
Outstanding Exposure	₹10.49B
Payments Collected	₹1.88B
Collection Rate	17.92%
Accounts with PTP	13,532
Accounts with Payments	16,934

Business Problem

CredResolve operates in technology-driven recovery management, where the effectiveness of collection depends on identifying the right accounts, applying suitable collection strategies, and converting customer engagement into successful payments.

The provided dataset contains information from different stages of the collections process, including accounts, borrowers, targeting, calls, communication activities, field visits, promises-to-pay, and payments.

However, these datasets contain operational challenges such as missing values, duplicate records, repeated identifiers, and inconsistent relationships between tables. Therefore, the first challenge is to establish reliable data before calculating business metrics.

The second challenge is to understand **where the outstanding financial exposure is concentrated, how collection activities are performing, and whether customer commitments such as promises-to-pay are translating into actual payments.**

The final challenge is to identify accounts that may require greater collection attention based on measurable business signals such as outstanding amount, delinquency, payment activity, and unresolved promises-to-pay.

Therefore, this project aims to transform the raw collections data into **clear, reliable, and actionable collection intelligence** that can support better prioritization and recovery decisions.

Collection Lifecycle

This is important because it shows the recruiter that you understand the **business process**, not just Python.

The analysis follows the collections lifecycle:

Account → Targeting → Collection Activity → Customer Engagement → Promise to Pay → Payment

Stage	What it represents
Account	Loan/account exposure, outstanding amount and delinquency
Targeting	Accounts selected for collection campaigns
Collection Activity	Calls, messages and field visits
Customer Engagement	Responses and collection outcomes
Promise to Pay	Customer commitment to make a payment
Payment	Actual amount successfully collected

Key Business Questions

1. **Portfolio Exposure:**
Where is the outstanding loan exposure concentrated?
2. **Collection Performance:**
Which risk segments or account groups have better or weaker collection rates?
3. **Collection Effectiveness:**
Does collection activity such as calls and field visits translate into higher payment outcomes?
4. **Promise-to-Pay Conversion:**
How many accounts with promises-to-pay eventually make payments?
5. **Delinquency & Risk:**
How does delinquency (DPD) relate to outstanding exposure and payment behaviour?
6. **Data Quality:**
What data-quality issues could affect the accuracy of collection KPIs and reporting?
7. **Collection Prioritization:**
Which accounts should be prioritized for further collection action based on exposure, delinquency, payment activity, and unresolved PTP?

Project Objectives

To analyze the CredResolve collections data and transform raw operational records into reliable, actionable insights that can support better collection prioritization and recovery decisions.

Specific Objectives

1. **Understand the data**
Examine the structure and relationships between the provided datasets.
2. **Improve data quality**
Identify missing values, duplicate records, inconsistent identifiers, and relationship issues before analysis.

3. **Build reliable analytical data**
Prepare account-level data by combining relevant information from accounts, collection activities, PTP, and payments.
4. **Measure collection performance**
Calculate key metrics such as outstanding exposure, payments collected, collection rate, PTP accounts, and payment activity.
5. **Perform SQL analysis**
Use SQL to answer practical business questions related to exposure, payments, collection activity, and account prioritization.
6. **Perform exploratory analysis**
Use Python and visualizations to identify patterns across risk, delinquency, collection activity, and payment behaviour.
7. **Analyze PTP and payment outcomes**
Examine whether promises-to-pay are associated with actual payment activity.
8. **Develop a collection-prioritization approach**
Identify accounts that may require greater attention using explainable business signals such as exposure, DPD, payment status, and unresolved PTP.
9. **Provide actionable recommendations**
Translate the analytical findings into practical recommendations for improving collection effectiveness.

Approach: Data Quality → Data Preparation → KPI Analysis → SQL Analysis → EDA → Prioritization
→ Business Recommendations

Dataset Overview

The CredResolve assignment provides a synthetic collections analytics dataset containing multiple relational tables representing different entities and events in the recovery management lifecycle.

The datasets cover borrower and account information, collection activities, communication channels, promises-to-pay, payments, campaigns, agent operations, complaints, and account status changes.

Since the data is distributed across multiple related tables, understanding the structure and relationships between these tables is an important first step before performing analysis.

Category	Dataset Examples	Purpose
Customer & Account	borrowers, accounts	Customer details, loan exposure and delinquency
Collection Activity	calls, call_attempts, call_dispositions	Call attempts and collection outcomes
Digital Communication	sms_events, whatsapp_events	Customer communication activity

Payment & Commitment	payments, promises_to_pay	Actual payments and customer commitments
Field Operations	field_visits	Physical collection activity
Targeting & Campaigns	daily_targeting, campaigns	Collection targeting and campaign information
Operations	agents, agent_sessions, vendor_telephony	Agent and telephony information
Customer Service	complaints	Customer complaints and resolution
Account History	account_status_history	Changes in account status
Reference	data_dictionary, README.md	Field definitions and dataset documentation

Dataset Size

The extracted dataset contains **19 files**, including **18 CSV datasets and one README documentation file**. The CSV tables have different row counts because they represent different business entities and event frequencies.

Dataset	Rows	Columns
Accounts	30,000	11
Borrowers	30,600	8
Calls	91,350	11
Call Attempts	120,000	9
Payments	25,500	9
Promises to Pay	18,000	9
Field Visits	25,000	10
Daily Targeting	45,000	7
SMS Events	45,000	8
WhatsApp Events	60,600	8
Complaints	8,000	9

Important Analytical Entities

The **account** is treated as the central analytical entity because collection decisions are ultimately made at the account level. Account information was connected with collection activities, PTP records, and payment outcomes using available identifiers.

Borrower → Account → Collection Activity → PTP → Payment

Data Quality Assessment & Cleaning

Why Data Quality Was Checked?

Before calculating business KPIs or performing analysis, the quality of the provided datasets was assessed. This was necessary because incorrect duplicates, missing values, inconsistent identifiers, or broken relationships between tables can lead to inaccurate collection metrics and business conclusions.

The assessment focused on missing values, exact duplicate records, duplicate business keys, identifier consistency, and relationships between major tables.

Missing Values

Missing-value analysis was performed across all datasets to identify columns requiring attention.

Dataset	Rows	Columns	Missing Cells	Missing %
Calls	91,350	11	1,827	0.18%
Borrowers	30,600	8	1,509	0.62%
WhatsApp Events	60,600	8	0	0.00%
Payments	25,500	9	382	0.17%

Missingness was relatively low in the major operational tables shown above. Missing values were therefore reviewed based on their business meaning rather than automatically replacing every missing value.

Duplicate Records

Exact duplicate records were identified in several event tables. These duplicates are important because repeated operational records can inflate activity counts and distort collection KPIs.

Dataset	Exact Duplicate Rows
Calls	1,271
Borrowers	600
WhatsApp Events	600
Payments	486

Exact duplicates were identified and considered during preparation of the analytical data. The raw source data was preserved, while duplicate records were handled in the analytical layer to avoid changing the original dataset.

Duplicate Business Keys

A second level of validation checked whether expected business identifiers were unique. Several tables contained repeated identifiers, indicating that a column that appears to be an identifier should not automatically be assumed to represent a unique record.

Dataset	Key	Rows	Unique Keys	Duplicate Key Rows
Borrowers	borrower_id	30,600	11,015	19,585
Agents	agent_id	30,000	1,000	29,000
Calls	call_id	91,350	90,000	1,350
Payments	payment_id	25,500	25,000	500
WhatsApp Events	whatsapp_event_id	60,600	60,000	600

These results show that identifier uniqueness cannot be assumed across every table. For example, the agent_id appears repeatedly because agents can participate in multiple operational records. Therefore, analytical aggregation must be performed at the correct business grain rather than simply counting identifier values.

Relationship Validation

Key relationships between major datasets were validated before creating account-level metrics. This helped identify whether transaction and activity records could be reliably connected to the central account table.

Validation	Result
Payments with account IDs not found in Accounts	0
Calls with account IDs not found in Accounts	0
Accounts with borrower IDs not found in Borrowers	2,913

Payments and calls had complete account-level relationships, allowing them to be safely linked to the accounts table for account-level analysis. However, **2,913 account records had borrower IDs that were not found in the borrowers table**. This relationship issue was retained as a data-quality finding rather than silently removing those accounts.

Data Quality Actions

Based on the assessment, the following principles were applied:

- Preserve the original raw data.

- Identify exact duplicates before calculating event-based metrics.
- Validate identifiers before using them as keys.
- Validate relationships between major tables.
- Avoid deleting records solely because a related table has missing information.
- Perform KPI calculations using validated analytical datasets

Data-quality assessment showed that the dataset intentionally contains real-world challenges such as duplicates, missing values and relationship inconsistencies. Addressing these issues before analysis improves the reliability of the resulting collection KPIs and business insights.

Data Preparation & Analytical Dataset

Preparation Approach

After assessing the quality of the raw datasets, the relevant tables were prepared for analysis. The objective was to create reliable account-level information without unnecessarily changing the original source data.

The **accounts table was used as the central reference**, and collection activity, promises-to-pay, and payment information were connected using the available account identifiers.

Dates and timestamps were standardized where required, numerical fields were reviewed for valid values, and duplicate records were considered before calculating activity and financial metrics.

Analytical Flow

Raw Tables → Data Quality Checks → Cleaning & Validation → Account-Level Aggregation → KPIs / SQL / EDA

Analytical Component	Main Data Used	Purpose
Portfolio Analysis	Accounts	Exposure, DPD and risk
Collection Activity	Calls, call attempts, field visits	Measure collection effort
PTP Analysis	Promises-to-pay	Measure customer commitments
Payment Analysis	Payments	Measure actual recovery
Targeting Analysis	Daily targeting, campaigns	Understand collection prioritization
Customer Analysis	Borrowers	Customer-level context

Account-Level Analytical View

To make the analysis easier to interpret, account-level measures were derived from the underlying event tables. This allowed multiple collection events and payment records to be summarized against each account.

Important account-level measures included:

- Outstanding amount
- DPD
- Risk segment
- Total payment amount
- Payment presence
- PTP presence
- Collection activity
- Collection priority

KPI Calculation

The main portfolio KPIs were calculated using validated account and payment information

Collection Rate

Collection Rate = Total Payments Collected ÷ Total Outstanding Exposure × 100

For this dataset:

Collection Rate = ₹1.88B ÷ ₹10.49B × 100 = 17.92%

Why Account-Level Analysis?

This is a nice little explanation that makes you sound like you understand the data:

Event tables such as calls, messages and payments can contain multiple records for the same account. Therefore, directly counting event rows can exaggerate the number of unique accounts or the amount of collection activity.

Aggregating relevant events at the **account level** provides a consistent analytical grain and makes it easier to compare collection effort, delinquency, PTP behaviour and payment outcomes.

The resulting analytical dataset provides a consistent foundation for KPI calculation, SQL analysis, exploratory visualization, and collection prioritization.

Collection KPIs & Exploratory Data Analysis

Portfolio-Level KPIs

The first stage of analysis establishes a portfolio-level view of exposure and recovery performance. These KPIs provide a baseline for understanding the scale of the collection challenge before examining individual segments and accounts.

KPI	Value
Total Outstanding Exposure	₹10.49B
Total Payments Collected	₹1.88B
Collection Rate	17.92%
Accounts with PTP	13,532
Accounts with Payments	16,934

The portfolio has approximately ₹10.49 billion in outstanding exposure against ₹1.88 billion in recorded payments. The resulting collection rate is 17.92%, indicating that a substantial portion of the outstanding portfolio remains unrecovered.

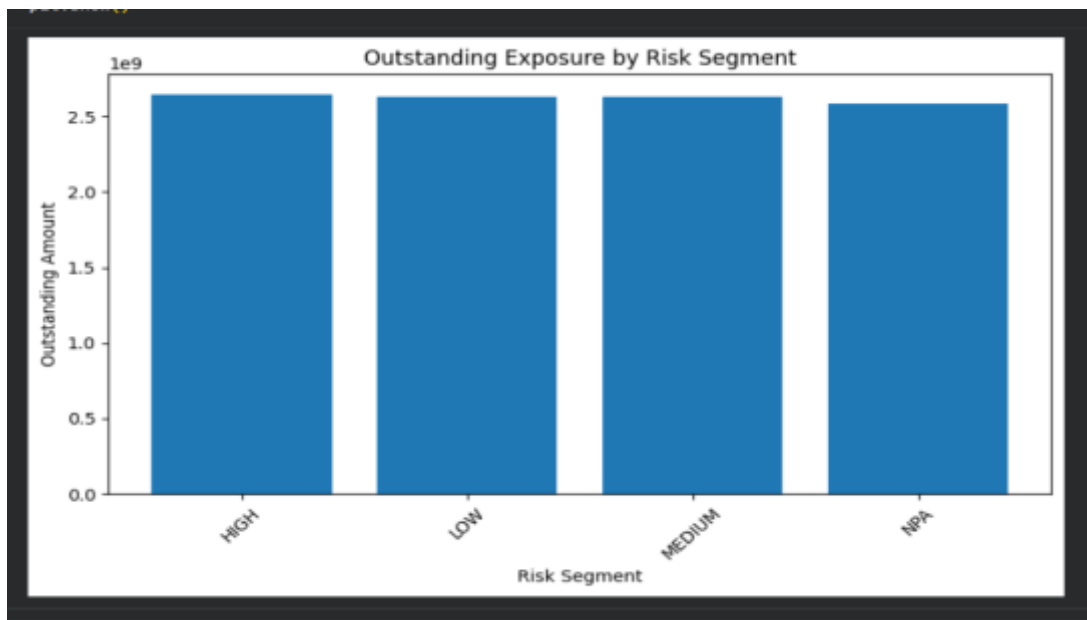
Risk Segment Analysis

Chart title:

Outstanding Exposure by Risk Segment

This analysis compares outstanding financial exposure across risk segments. It helps identify where the largest amount of recoverable exposure is concentrated.

Business relevance: Risk segmentation can help collection teams allocate resources according to both account risk and financial impact.



Collection Rate by Risk Segment

Insert your **collection rate by risk segment** chart.

Title:

Collection Rate by Risk Segment

Collection rate is compared across risk segments to identify differences in recovery performance. A segment with a lower collection rate may require further investigation into customer behaviour, collection strategy, or channel effectiveness.

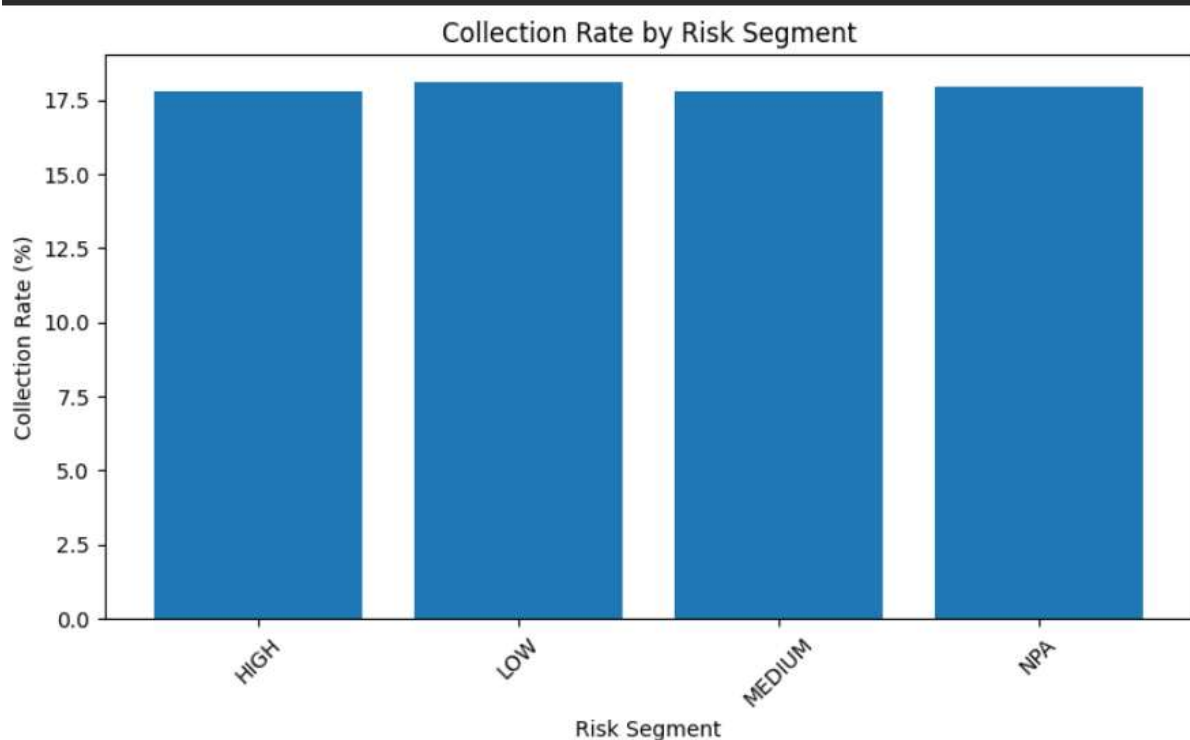
Business relevance:

Segment-level collection rates provide more insight than a portfolio-level average because they reveal where recovery performance differs across the portfolio.

Outstanding Exposure vs Payments

This visualization compares the amount outstanding with the amount collected for individual accounts. It helps identify accounts with substantial exposure but relatively limited payment activity.

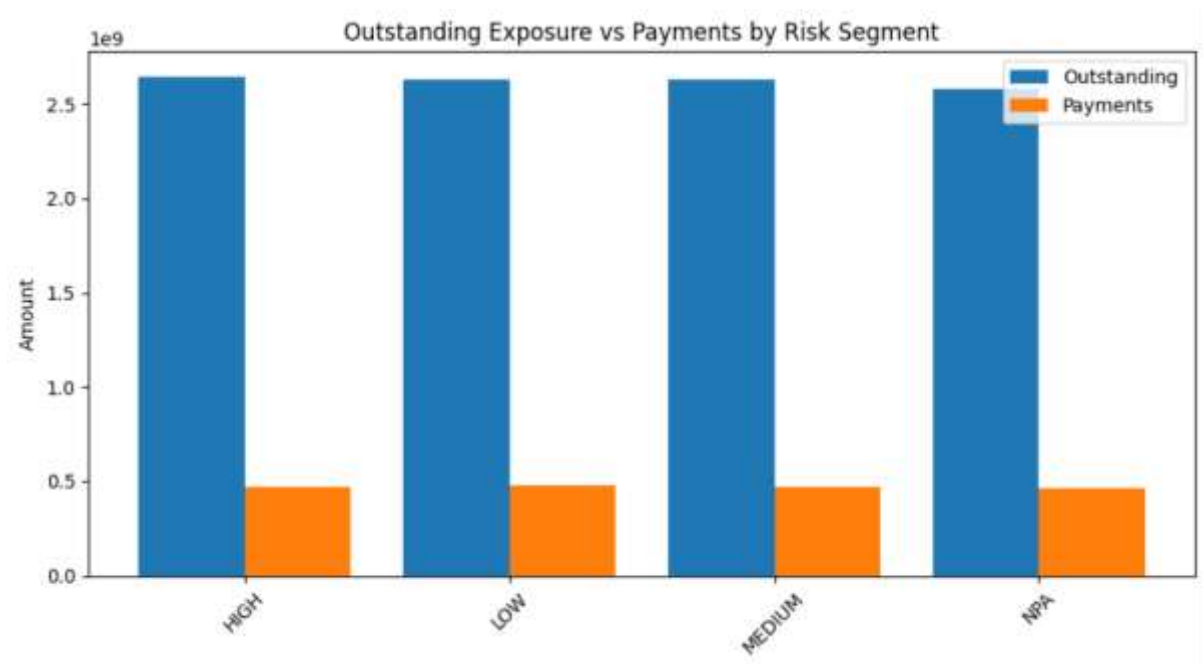
Business relevance: High-exposure accounts with low or no payments may represent important opportunities for targeted collection action.



Collection Activity vs Payment Outcome

Collection activity is compared with payment outcomes to understand whether increased operational effort is associated with higher recorded recovery.

Business relevance: Collection teams should evaluate activity based on recovery outcomes rather than using contact volume alone as a measure of effectiveness.

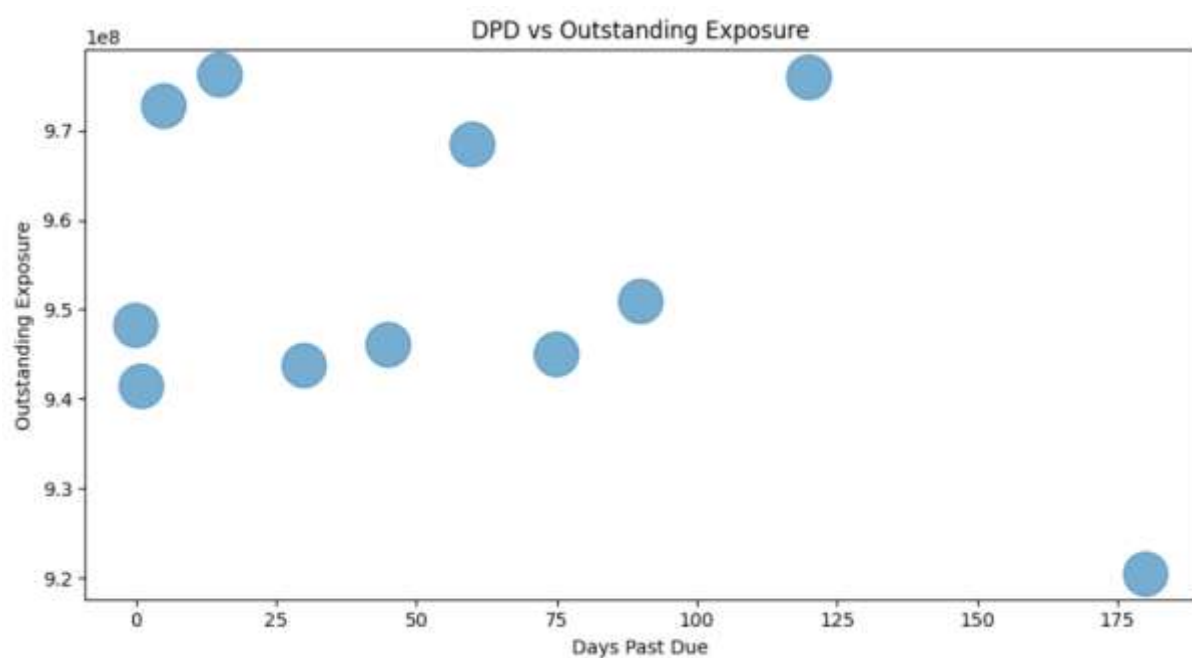


DPD vs Outstanding Exposure

Delinquency (DPD) vs Outstanding Exposure

This analysis examines the relationship between days past due (DPD) and outstanding exposure. Combining delinquency with financial exposure provides a more useful view of potential collection priority than either measure alone.

Business relevance: Accounts with both high delinquency and high outstanding exposure may warrant stronger collection attention.



PTP & Payment Analysis

Promise-to-Pay and Payment Behaviour

Promises-to-pay represent an important stage in the collections lifecycle because they indicate a customer's stated intention to make a payment. However, a promise alone does not represent recovered cash.

Therefore, PTP records were analyzed alongside payment records to distinguish between accounts that made commitments and accounts that subsequently had recorded payments.

13,532 accounts with PTP

Versus

16,934 accounts with payments

These figures show that PTP activity and payment activity represent different stages of the recovery process and should be analyzed together rather than treated as equivalent outcomes.

EDA Summary

The exploratory analysis provides three important perspectives:

- 1. Financial perspective:** Where is outstanding exposure concentrated?
- 2. Operational perspective:** How much collection activity is being performed?
- 3. Recovery perspective:** How are PTP and collection activity associated with actual payments?

These perspectives are combined in the next sections to develop an explainable collection-prioritization approach.

SQL Analysis & Collection Prioritization

SQL Business Analysis

SQL was used to answer business-oriented questions across the relational collections data. The objective was not simply to demonstrate SQL syntax, but to use SQL for extracting information that can support collection decisions.

Business questions answered using SQL

#	Business Question	SQL Approach
1	Which accounts have the highest outstanding exposure?	ORDER BY + filtering
2	Which risk segments contain the highest exposure?	GROUP BY + aggregation
3	Which accounts have payments recorded?	JOIN + aggregation
4	Which accounts have PTP but no recorded payment?	LEFT JOIN + filtering

5	Which accounts have high exposure with low/no payment activity?	Aggregation + conditional logic
6	How does collection activity vary across accounts?	GROUP BY + counts

High-Exposure Accounts

SELECT

account_id,
outstanding_amount,
dpd,
risk_segment

FROM accounts

ORDER BY outstanding_amount DESC

LIMIT 10;

Purpose: Identify accounts with the highest outstanding financial exposure.

	risk_segment	accounts	outstanding_exposure
0	HIGH	7552	2.646183e+09
1	LOW	7513	2.633311e+09
2	MEDIUM	7533	2.628179e+09
3	NPA	7402	2.581362e+09

PTP Without Recorded Payment

SELECT

p.account_id,
p.promised_amount,
p.promised_date

FROM promises_to_pay p

LEFT JOIN payments pay

ON p.account_id = pay.account_id

WHERE pay.account_id IS NULL;

	account_id	risk_segment	dpd	outstanding_amount	ptp_count	promised_amount	total_payment
0	ACC0004015	NPA	15	167410.59	6.0	361703.11	0.0
1	ACC0009085	LOW	15	301733.12	4.0	330948.07	0.0
2	ACC0009182	NPA	1	508003.31	4.0	328352.49	0.0
3	ACC0013947	LOW	0	675964.88	4.0	308559.68	0.0
4	ACC0019034	HIGH	90	462734.89	4.0	303513.10	0.0
5	ACC0018958	MEDIUM	90	387984.76	6.0	301925.17	0.0
6	ACC0001250	MEDIUM	120	515758.26	4.0	297499.91	0.0
7	ACC0008696	HIGH	90	359025.39	3.0	282620.75	0.0
8	ACC0019952	HIGH	1	667362.67	4.0	280797.55	0.0
9	ACC0013330	LOW	120	400704.02	4.0	278405.88	0.0
10	ACC0027437	MEDIUM	120	573034.06	3.0	271085.49	0.0
11	ACC0024994	MEDIUM	120	174896.54	4.0	269632.06	0.0
12	ACC0007720	LOW	30	96686.76	3.0	268477.73	0.0

Purpose: Identify accounts where a promise-to-pay exists but no corresponding payment record is present.

Collection Prioritization Framework

ased on the business objective, an explainable rule-based prioritization framework was developed to identify accounts that may require greater collection attention.

Instead of relying on a complex machine-learning model, the framework uses understandable business signals that a collection team can directly interpret.

Signals considered	
Signal	Why it matters
Outstanding Amount	Represents financial exposure
DPD	Indicates delinquency severity
Payment Activity	Shows whether recovery has already occurred
PTP Status	Indicates customer commitment
Unresolved PTP	May indicate a follow-up opportunity

Priority Logic

Use this simple framework:

● High Priority

Accounts with combinations such as:

- High outstanding amount
- High DPD
- No recorded payment
- PTP exists but payment is not recorded

● Medium Priority

Accounts showing:

- Moderate exposure
- Moderate delinquency
- Some collection/payment activity
- Partial recovery

● Lower Priority

Accounts showing:

- Lower exposure
- Lower delinquency
- Successful payment activity
- No major unresolved collection signal

These accounts may warrant higher collection attention based on the selected business signals.

Why an Explainable Approach?

The prioritization framework is intentionally explainable. Collection teams should be able to understand why an account receives a higher priority rather than relying on an opaque score.

This approach also provides a foundation for future predictive modelling. Once sufficient historical outcome data is validated, the rule-based framework could be compared with a probability-of-payment or recovery prediction model.

Key Analytical Insights

Key Insights

Based on the analysis:

1. Significant financial exposure remains unrecovered

The portfolio contains approximately **₹10.49B in outstanding exposure** compared with **₹1.88B in recorded payments**, resulting in a **17.92% collection rate**.

2. Collection activity should be evaluated by outcomes

High activity volume alone does not necessarily represent collection effectiveness. Operational effort should be evaluated alongside payment outcomes.

3. PTP should be monitored as a separate recovery stage

A promise-to-pay represents customer commitment, while a payment represents actual recovery. Therefore, PTP follow-up should be monitored separately from payment performance.

4. Data quality can directly affect collection reporting

Duplicate records and inconsistent relationships can inflate activity counts or cause incorrect joins. Data-quality controls should therefore be part of the reporting process.

5. Financial exposure and delinquency should be considered together

Prioritizing accounts only by DPD or only by outstanding amount can miss important business opportunities. Combining both signals provides a more useful basis for collection prioritization

Overall, the analysis moves from descriptive reporting toward decision support by combining financial exposure, delinquency, collection activity, PTP behaviour, payment outcomes, and data quality into an explainable prioritization framework

Business Recommendations

Based on the analysis, the following actions can support more effective collection management.

Prioritize High-Impact Accounts

Collection teams can prioritize accounts using a combination of **outstanding exposure, DPD, payment activity, and unresolved PTP status** rather than relying on a single factor.

Why: This helps focus limited collection resources on accounts with potentially higher financial impact.

Strengthen PTP Follow-Up

Accounts with a promise-to-pay but without a corresponding payment should be monitored through a dedicated follow-up process.

Why: A PTP represents customer commitment, but the actual recovery outcome is the payment.

Evaluate Collection Channels by Outcomes

Calls, digital communication, and field visits should be evaluated based on their relationship with payment outcomes rather than activity volume alone.

Why: A high number of contacts does not automatically indicate effective recovery.

Introduce Data-Quality Monitoring

Duplicate records, repeated identifiers, missing values, and relationship inconsistencies should be monitored as part of the regular reporting pipeline.

Why: Poor data quality can directly affect collection KPIs and operational decisions.

Use Risk + Financial Exposure Together

Collection prioritization should consider both **delinquency severity and financial exposure**.

Why: An account with high DPD but very low exposure may have a different business priority from an account with high DPD and substantial outstanding exposure

Future Analytical Opportunities

The current analysis provides a foundation for further collection intelligence. Future improvements could include:

- **Probability-of-payment prediction**
- **Expected recovery amount prediction**
- **Channel effectiveness modelling**
- **Agent performance benchmarking**
- **Customer segmentation**
- **Early-warning models for missed PTP**
- **Automated collection-priority dashboards**
- **Machine-learning based recovery scoring**

These approaches should be implemented only after validating historical outcomes, data quality, and potential data leakage.

Limitations

1. The provided dataset is a **synthetic dataset**, so patterns may not fully represent real customer behaviour.
2. The current prioritization approach is **rule-based and explainable**, rather than a predictive machine-learning model.
3. A recorded collection activity does not necessarily establish causation between the activity and a payment.

4. Some data-quality issues, such as unmatched borrower relationships, limit the interpretation of certain customer-level analyses.
5. Further historical information would be required to develop and validate a robust predictive recovery model.

6. Conclusion

- This project transformed a multi-table collections dataset into a structured analytical view of the recovery lifecycle.
- The analysis began with data-quality assessment and relationship validation, followed by portfolio KPI calculation, exploratory analysis, SQL-based business analysis, and account-level collection prioritization.
- The portfolio contains approximately ₹10.49B in outstanding exposure and ₹1.88B in recorded payments, resulting in a 17.92% collection rate. The analysis also identified important data-quality considerations, including duplicate records and relationship inconsistencies that need to be considered when building reliable collection reports.
- Rather than treating collection activity, PTP, and payments as isolated metrics, the project connects them across the recovery lifecycle to provide a more complete view of collection performance.
- The resulting prioritization framework provides an interpretable starting point for collection decision-making and can be extended in the future into predictive recovery and payment-probability models.
- Overall, the project demonstrates how data quality, SQL, Python analysis, visualization, and business reasoning can be combined to turn operational collection data into actionable intelligence.